# Introduction

# Introduction I

The scholarly literature on any active topic grows faster than any individual can read. A researcher entering a field needs to know how large it is, how fast it is growing, what sub-areas compose it, which works anchor its citation structure, what language and concepts recur, and which claims the field actually tests. Answering those questions by hand is slow, unrepeatable, and binds no number to evidence. Systematic reviews, while the gold standard for evidence synthesis, are labour-intensive and become stale between updates — the Cochrane review cycle, for instance, targets a two-year refresh interval that many fields outpace [@page 2021prisma]. Bibliometric dashboards (Scopus, Web of Science, Dimensions) offer breadth but no reproducible link from a reported statistic to a regenerable artifact, and they do not frame or test domain-specific hypotheses.

# Introduction II

This project is a **configurable, reproducible meta-analysis template**. It takes one search term and produces a quantitative portrait of that term's literature, with every reported number traceable to a committed artifact and regenerable by re-running the pipeline. The bundled instance targets **Modafinil**, a wakefulness-promoting agent with a large, multi-disciplinary literature spanning clinical sleep medicine, cognitive neuroscience, pharmacology, psychiatry, and safety research; pointing the configuration at a different term re-targets the whole analysis with no code change.

# Research Questions I

The pipeline is designed around four research questions (RQs) that a researcher entering the field would ask:

1. **RQ1 — Field size and growth.** How large is the literature on Modafinil, how fast is it growing, and when did it peak? The corpus of $N = 2334$ records spanning 2000–2026 answers this directly, with a compound annual growth rate of 5.48% and a peak in 2025.
2. **RQ2 — Subfield composition.** What sub-areas compose the literature, and what is their relative weight? A configurable 6-bucket taxonomy (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience) classifies every record, with **Clinical Sleep** the largest bucket at 63.0%.

# Research Questions II

3. **RQ3 — Topical and linguistic structure.** What language and concepts recur, and what latent topics cross-cut the keyword taxonomy? TF-IDF over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics. The top vocabulary terms are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant.
4. **RQ4 — Citation geometry and evidence landscape.** Which works anchor the citation structure, how self-contained is the retrieved slice, and which claims does the field test? The citation network of 2234 nodes and 8,623 edges (22.4% reference resolution rate) exposes hubs, authorities, and communities, while 6 configured hypotheses frame the evidence landscape.

# Contributions I

The pipeline contributes an end-to-end, domain-agnostic workflow:

1. **Multiple-engine retrieval with graceful degradation.** Records are gathered from 10 independent engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv). An engine with no API key or no network reports a *skipped* status; the run completes from whatever engines remain plus a committed offline corpus. Each new retrieval persists per-engine outcome and count provenance in `output/data/retrieval_report.json`; the merged corpus alone is never used to infer which engine supplied a record.
2. **Record de-duplication.** Heterogeneous records are merged by a canonical identifier hierarchy, keeping the most complete version of each work. Of 2334 retrieved records, 2260 carry DOIs, 923 carry OpenAlex IDs, and 1 carry arXiv IDs.

# Contributions II

3. **Descriptive and bibliometric analysis.** Counts by year, venue, and author; growth metrics (CAGR 5.48%, doubling time 9.2 years); a configurable 6-bucket subfield classification; topic models; and a citation network with 1416 communities.

4. **Language, entity, and embedding analysis.** Keyphrase and entity extraction and offline deterministic document embeddings over titles, abstracts, and full text. The TF-IDF vocabulary of 500 features captures the lexical landscape.

5. **Optional hypothesis evidence.** An LLM-gated knowledge-graph stage scores the 6 configured hypotheses explored against the corpus.

Because the writing itself is token-injected from configuration and pipeline outputs, the manuscript is part of the reproducible artifact rather than a separate hand-authored narrative. Every number, table, and figure reference in this document traces to a committed, regenerable file under `output/`.